

LumiTrack FSOC SIL Testbed

Coarse PAT Simulation Experiment Report

Scenario: Target Occlusion & Loss Test

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SIH Problem Statement: 26169 — AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile FSOC Terminals

Performance Summary

Metric	Value	Threshold	Assessment
Initial Acquisition Time	11.77 s	< 3.0 s	✗ FAIL
Lock Retention Rate	16.2%	≥ 80.0%	✗ FAIL
Average Tracking Error	22.31 px / 2.091°	< 2.0°	✗ FAIL
Max Tracking Error	151.32 px	—	—
Effective Frame Rate	9.9 FPS	≥ 25.0 FPS	✗ FAIL
Avg Processing Latency	85.37 ms	< 33 ms	—
Target Loss Events	4	—	—
Successful Re-acquisitions	2	—	—
Frames Processed	560	—	—

Telemetry Charts



FSOC PAT Coarse Alignment Assessment

✗ COARSE ALIGNMENT CRITERIA NOT MET

One or more FSOC PAT coarse alignment performance thresholds were not met in this scenario.
Recommended actions: tune PID gains ($K_p/K_i/K_d$), increase camera FOV, or reduce disturbance levels.